

Jiaheng Wang

HIIT Postdoctoral Fellow



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Employment

2026–Present	HIIT postdoctoral fellow	University of Helsinki
2024–2026	Postdoctoral researcher	University of Regensburg
2023–2024	Postdoctoral researcher	University of Edinburgh

Education

2020–2023	Ph.D. Thesis title: <i>Algorithms and complexity for approximately counting hypergraph colourings and related problems.</i> Advisor: Heng Guo.	University of Edinburgh
2016–2020	B.Sc. <i>summa cum laude</i> in Computer Science (Turing Class).	Peking University

Research topics

- Algorithms and complexity, with a focus on **counting problems** and their connection with other topics, including
 - randomness and approximability in computation;
 - fine-grained, parameterised and algebraic complexity;
 - applications in statistical physics, computational geometry, learning theory, etc.

Publications & preprints

- **Planar Perfect Matching Counting is as Hard as Determinants.**
Radu Curticapean, Jiaheng Wang.
preprint
arXiv: 2606.03975
- **Partition Rank and Algebraic Circuit Lower Bounds.**
Cornelius Brand, Petteri Kaski, Jiaheng Wang.
ESA 2026, to appear
arXiv: 2607.02241

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- **Beyond Bilinear Complexity: What Works and What Breaks with Many Modes?**
Cornelius Brand, Radu Curticapean, Petteri Kaski, Baitian Li, Ian Orzel, Tim Seppelt, Jiaheng Wang.
[CCC 2026](#)
arXiv: 2602.11975
 - **Sink-free orientations: a local sampler with applications.**
Konrad Anand, Graham Freifeld, Heng Guo, Chunyang Wang, Jiaheng Wang.
[RANDOM 2025](#)
arXiv: 2502.05877
 - **Can you link up with treewidth?**
Radu Curticapean, Simon Döring, Daniel Neuen, Jiaheng Wang.
[Journal of Computer and System Sciences](#), Volume 162, Article 103835, 1–25, 2026
[STACS 2025](#)
arXiv: 2410.02606
 - **Rapid mixing of the flip chain over non-crossing spanning trees.**
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, Mark Jerrum, Jiaheng Wang.
[SoCG 2025](#)
arXiv: 2409.07892
 - **The complexity of computing fermionants and flow-like structures in graphs, modulo p .**
Isja Mannens, Jiaheng Wang.
submitted
 - **Approximate counting for spin systems in sub-quadratic time.**
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, Jiaheng Wang.
[TheoretiCS](#), Volume 4 (2025), Article 3, 1–27
[ICALP 2024](#)
arXiv: 2306.14867
 - **Inapproximability of counting independent sets in linear hypergraphs.**
Guoliang Qiu, Jiaheng Wang.
[Information Processing Letters](#), Volume 184, Article 106448, 1–6, 2024
arXiv: 2212.03072
 - **Towards derandomising Markov chain Monte Carlo.**
Weiming Feng, Heng Guo, Chunyang Wang, Jiaheng Wang, Yitong Yin.
[SIAM Journal on Computing](#), Vol. 54, Iss. 3, pp. 775–813, 2025
[FOCS 2023](#)
arXiv: 2211.03487
 - **A simple polynomial-time approximation algorithm for the total variation distance between two product distributions.**
Weiming Feng, Heng Guo, Mark Jerrum, Jiaheng Wang.
[TheoretiCS](#), Volume 2 (2023), Article 8, 1–7
[SOSA 2023](#)
arXiv: 2208.00740
 - **Swendsen–Wang dynamics for the ferromagnetic Ising model with external fields.**
Weiming Feng, Heng Guo, Jiaheng Wang.
[Information and Computation](#), Volume 294, Article 105066, 1–34, 2023
arXiv: 2205.01985

- **Improved bounds for randomly colouring simple hypergraphs.**
Weiming Feng, Heng Guo, Mark Jerrum, Jiaheng Wang.
RANDOM 2022
arXiv: 2202.05554
- **Inapproximability of counting hypergraph colourings.**
Andreas Galanis, Heng Guo, Jiaheng Wang.
ACM Transactions on Computation Theory, 14(3-4):10, pp. 1-33, 2022
arXiv: 2107.05486
- **On the degree of Boolean functions as polynomials over $\mathbb{Z}_m$.**
Xiaoming Sun, Yuan Sun, Jiaheng Wang, Kewen Wu, Zhiyu Xia, Yufan Zheng.
ICALP 2020
arXiv: 1910.12458

Awards

2020	Informatics Global PhD Scholarship (3.5 years)	University of Edinburgh
2016–2020	4 awards/scholarships during undergraduate study	Peking University

Services

- Programme committee member:
FSTTCS 2025
- Layout editor of TheoretiCS since 2025
Voluntary. This is a Diamond Open Access journal published by the TheoretiCS Foundation e.V., a German not-for-profit organization. Please consider supporting it!
- External reviewer at conferences:
FSTTCS 2026, ISAAC 2026, ESA 2026, ICALP 2026, STOC 2026, STACS 2026, SODA 2026, MFCS 2025, ESA 2025, CIAC 2025, APPROX/RANDOM 2024, FOCS 2024, ICALP 2021, SODA 2021
- Journal referee:
Theoretical Computer Science
- Student organizer of SAGT 2018
organizing volunteers, getting involved in press, etc.

Teaching

University of Regensburg:		
2025 Winter	76388 Randomized Algorithms	Durchführender
2024 Winter	70101 Complexity Theory	Durchführender
University of Edinburgh:		
2022–2023	INFR08026 Introduction to Algorithms and Data Structures	Teaching Assistant/Tutor
2022 Autumn	INFR11201 Randomized Algorithms	Tutor

2021–2022	INFR08026 Introduction to Algorithms and Data Structures	Teaching Assistant/Tutor
	● Peking University:	
2020 Spring	04834010 Randomized Algorithms	Teaching Assistant
2020 Spring	04833440 Introduction to the Theory of Computation	Teaching Assistant
2019 Fall	04833040/04832363 Introduction to Computer Systems	Teaching Assistant/Tutor
2019 Spring	04833440 Introduction to the Theory of Computation	Teaching Assistant
2018 Fall	04833040/04832363 Introduction to Computer Systems	Teaching Assistant/Tutor